

Mini Project

Tube manipulator robot

Team members

Last name	First name	Last name	First name	Students
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Prepared by:

Name	Signature	Date	Supervisor
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Controlled by:

Name	Signature	Date	First team member
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Approved by:

Name	Signature	Date	Second team member
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Tests results confirmed by:

Name	Signature	Date	First team member
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Tests results confirmed by:

Name	Signature	Date	Second team member
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Tests results approved by:

Name	Signature	Date	Supervisor
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Degree Programme
Systems Engineering

Sion, Mai 12, 2026

Performance Qualification

1. Time and Date.

PQ ID	Timestamp for file
1.1	Tested with PickTimeStamp.csv or PlaceTimeStamp.csv No test here
PQ ID	Timestamp for record
1.2	Tested in PickTimeStamp.csv or PlaceTimeStamp.csv No test here
PQ ID	Timezone
1.3	Tested with PickTimeStamp.csv or PlaceTimeStamp.csv No test here

2. Simplified Control

PQ ID	Layout
2.1	The simplified control address is : http://hostname:1880/.....(page name) - The page has a notebook layout (yes/no) - The page has a Clear button (yes/no) - The page has a Reset button (yes/no) - The page has a Start button (yes/no) - The page has a Stop button (yes/no) - The page has a Pick button (yes/no) - The page has a Place button (yes/no) - The page has a Backup button (yes/no)
2.2	-The page has a nest field (yes/no) -The page has a velocity field (yes/no) -The page has an acceleration field (yes/no)

3. Machine Startup

PQ ID	Startup
3.1	The machine is set in state Aborted by opening the door. The door is closed The state of the machine is (check Siemens UI): - The operator presses the clear button, the state on the Siemens UI is : - The operator presses the reset button, the state on the Siemens UI is : - The operator presses the start button, the state on the Siemens UI is :
3.2	The machine is in Execute State. - The operator presses the stop button, the state on the Siemens UI is :

4. Pick and Place Sequence

PQ ID	Pick sequence
4.1	The machine is set in execute state. The state of the machine is (check Siemens UI): The operator set the velocity in 0.1 [m/s]: velocity in the field is[m/s] The operator set the acceleration in 2 [m/s²]: acceleration in the field is[m/s²] The operator selects nest N°:(nest number) The operator presses the button Pick.

	- The robot moves above to nest N° - The robot moves down (yes/no) - The robot moves up (yes/no) - The gripper is closed (yes/no)
PQ ID	Place sequence
4.2	The machine is set in execute state. The state of the machine is (check Siemens UI): The operator set the velocity in 0.2 [m/s] : velocity in the field is[m/s] The operator set the acceleration in 1.5 [m/s²]: acceleration in the field is[m/s²] The operator selects nest N°:(nest number) The operator presses the button Place. - The robot moves above to nest N° - The robot moves down (yes/no)..... - The robot moves up (yes/no)..... - The gripper is open (yes/no).....

5. Log events

URS ID	Log events and buttons
5.1	There is a logTubeEvent.json in the folder (enter path if any):..... There is a logAlarmsAndWarning.json in the folder (enter path if any):..... The test engineer deletes the logTubeEvent.json and logAlarmsAndWarning if any. The machine is set in aborted state by opening the door. The test engineer presses the following sequence of buttons. - Clear (wait stopped) - Reset (wait idle) - Start (wait execute) - Wait 10 seconds using a chronometer - Set velocity to 0.15 [m/s] - Set acceleration to 1.5 [m/s²]: - Pick - Wait 10 seconds using a chronometer - Place - Wait 10 seconds using a chronometer - Stop (wait stopped)
5.2	The test engineer makes a copy paste of: logTubeEvent.json to logTubeEvent - Copy.json Done : (yes/no) The logTubeEvent - Copy.json shall be provided with the report.
5.3	The test engineer make a copy paste of: logAlarmsAndWarning.json to logAlarmsAndWarning - Copy.json Done : (yes/no)...

	The logAlarmsAndWarning - Copy.json shall be provided with the report.
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6. Trace pick and place

PQ ID	PickTimeStamp.csv
6.1	There is a PickTimeStamp.csv file in the folder: - Folder name is:..... - Pickxxxx.csv complete name is:..... (select one if many)
6.2	The Pickxxxx.csv shall be provided with the report.
PQ ID	PlaceTimeStamp.csv
6.3	There is a PlaceTimeStamp.csv file in the folder: - Folder name is:..... - Placexxxx.csv complete name is:..... (select one if many)
6.4	The Placexxxx.csv shall be provided with the report.

7. Backup parameters

PQ ID	Backup
7.1	There is a CmdTimeStamp.xml file in the folder: - Folder name is:..... - Cmdxxxx.csv complete name is:..... (select one if many)
7.2	The test engineer presses the Backup button
7.3	The last version of Cmdxxxx.xml shall be provided with the report.

8. Version

URS ID	Description
8.1	A file with versions of software, including palettes is available (yes/no) ?.....

9. Archiving

URS ID	Description
9.1	Deadline and checkup of documents to provide read (yes/no) ?.....

Checkup

Following documents shall be provided in a zip document in Moodle. Deadline: Tuesday, May 26 at 8 a.m.

- o logTubeEvent - Copy.json
- o logAlarmsAndWarning - Copy.json
- o a PickTimeStamp.csv file
- o a PlaceTimeStamp.csv file
- o a CmdTimeStamp.xml file
- o summary of about two pages of what you have done.
- o The list of versions for software.